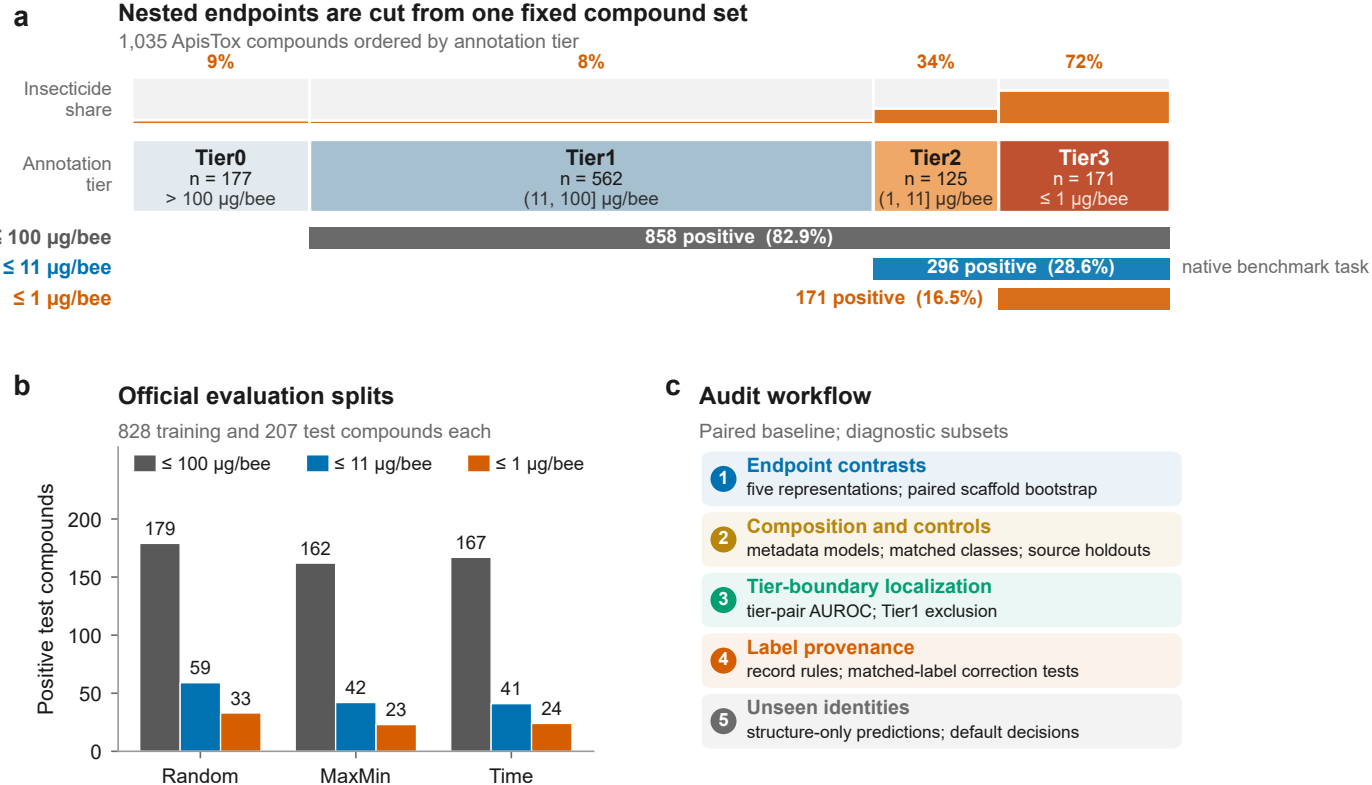

a
Nested endpoints are cut from one fixed compound set
1,035 ApisTox compounds ordered by annotation tier
9%
8%
34%
72%
Insecticide share
Annotation tier
Tier0
n = 177
> 100 µg/bee
Tier1
n = 562
(11, 100] µg/bee
Tier2
n = 125
(1, 11] µg/bee
Tier3
n = 171
≤ 1 µg/bee
≤ 100 µg/bee
858 positive (82.9%)
≤ 11 µg/bee
296 positive (28.6%)
native benchmark task
≤ 1 µg/bee
171 positive (16.5%)
b
Official evaluation splits
828 training and 207 test compounds each
≤ 100 µg/bee
≤ 11 µg/bee
≤ 1 µg/bee
Positive test compounds
0
50
100
150
200
179
59
33
162
42
23
167
41
24
Random
MaxMin
Time
c
Audit workflow
Paired baseline; diagnostic subsets
1
Endpoint contrasts
five representations; paired scaffold bootstrap
2
Composition and controls
metadata models; matched classes; source holdouts
3
Tier-boundary localization
tier-pair AUROC; Tier1 exclusion
4
Label provenance
record rules; matched-label correction tests
5
Unseen identities
structure-only predictions; default decisions